

Bhaumik Bhalara

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SUMMARY

Aspiring Electrical and Electronics Engineer with a strong foundation in hardware design and development, proficient in Verilog, Python, and FPGA technologies. Eager to leverage technical expertise in a dynamic environment focused on innovative hardware solutions and continuous improvement.

EXPERIENCE

Intern

Essar Power Hazira LTD, Surat

Jan 2022 - Mar 2022

- Gained hands-on experience with control systems for boiler-actuated power generation, enhancing operational efficiency by 20%.
- Conducted in-depth research on link mechanisms designed to transport waste loads from boilers, contributing to a 15% reduction in waste processing time.
- Monitored and recorded gauge and probe data, ensuring safety compliance through active participation in seminars, resulting in zero safety incidents during the internship.

Intern

3D Cubic, Surat

Aug 2021 - Oct 2021

- Completed an intensive internship focused on understanding and operating 3D printing technology, leading to a 30% improvement in production speed of prototypes.
- Showcased hands-on expertise by designing and fabricating a 2 cm x 2 cm x 2 cm polyamide cube using a 3D printer, demonstrating precision and attention to detail in manufacturing processes.

EDUCATION

Master of Science

California State University, Sacramento, Sacramento, CA, USA

Present

GPA: 3.23/4

Bachelor of Engineering

Sarvajani College of Engineering and Technology, Gujarat, India

May 2022

GPA: 3.30/4

SKILLS

Verilog, Python, FPGA Hardware Design, Control Systems, 3D Printing, System-Level Analysis, MATLAB, P-spice, Signal Generators, Oscilloscopes, DMM, CAD, Primetime, Quartus II IDE, Simplify3D, MobaXterm, G-Code, PCI-E, Intel DE-10 Lite FPGA Board

PROJECTS

64-bit PCI to SRAM Memory Card

Aug 2023 - Dec 2023

- Developed SRAM with detailed block and state diagrams for PCI to SRAM transactions, achieving a 95% accuracy in simulated environments.
- Prepared timing diagrams for both 64-64 bit and 32-64-bit bus transactions, optimizing data transfer efficiency by 25%.

Vending Machine System

Aug 2023 - Dec 2023

- Integrated finite state machines, including idle, selection, payment, dispensing, and reset states, improving user interaction time by 40%.
- Tested and verified the design using Verilog testbenches and simulation environments, successfully validating hardware on an Intel DE-10 Lite FPGA board.

6-Bit Segmented DAC

Jan 2024 - May 2024

- Executed the implementation of DAC divided into LSB and MSB segments, achieving expected results for both theoretical and simulated values of 12V with less than 5% error margin.
- Optimized design by adding two bits to reduce glitch noise, enhancing overall signal fidelity.

STA for Microprocessor Netlist

Aug 2024 - Dec 2024

- Executed design constraints script to generate comprehensive setup and hold timing reports, identifying and resolving 10+ timing violations to enhance design performance by 30%.
- Optimized slack through detailed analysis, improving overall design efficiency and reliability.

4-Bit Ripple Carry Adder

Aug 2024 - Dec 2024

- Implemented Verilog module and Perl script to verify the functionality of the adder, achieving a 100% success rate in functional tests.
- Generated timing reports by applying design constraints effectively, utilizing switches for static timing analysis, ensuring design robustness.